

the battery of course. High quality LCD screens will have bright, accurate colours and with visibility from just about any angle.

AMOLED

AMOLED (Active-matrix organic light-emitting diode) is a screen technology based on organic compounds that offers remarkably better image quality along with potentially very low power usage.

Till OLED screens came into the picture, all display technologies needed a backlight. This was the reason for their thickness, and each pixel would only control the colour shade. This also meant that the backlight would remain on and this 100% of the screen would be on. Unlike LCD screens, AMOLED displays don't need a backlight – each pixel produces its own light – so phones using them can potentially be thinner. Also, since each pixel can be turned off on its own this means that certain portions of the screen which would be displaying black colour could be turned off completely, this is what most people term as true black, i.e. absence of any light.

It also means that a mostly black screen will use very little electricity, and the colour reproduction is much better as opposed to TFT where dark regions would be reproduced as dark grey rather than true black.

This being said, AMOLED screens are rather expensive to produce in terms of cost and time, hence, you'll find these screens only in the upper mid-range and top-end products. Another drawback is that on a subpixel level, the arrangement of the subpixel is rather different as compared to LCD, thus there will be a slight distortion when compared to an LCD. High-end LCD screens which are found on certain top-end phones are capable of outdoing the colour reproduction capability of AMOLED screens.

Super AMOLED

An improvement over traditional AMOLED screens, Super AMOLED screens have superior brightness and colour reproduction. What is primarily different is that these screens have the capacitive touchscreen included in the manufacturing process and do not use a digitizer overlay as is the case in other screens.

Super AMOLED Plus

The subpixel arrangement of AMOLED screens led to a slightly distorted image being produced. With the advent of Super AMOLED Plus, this was fixed as the subpixel arrangement was modified to resemble that of an LCD, thus ensuring a cleaner image.